DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Semester Examination – Summer 2023

Course: B. Tech. **Branch: Civil Engineering** **Semester: II**

Subject Code & Name: BTES205/BTES205E Energy and Environment Engg.

Max Marks: 60 **Date: 21/7/2023** **Duration: 3 Hrs.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve the following.**		**12**
A)	Explain the working of a Hydro electric power plant with neat diagram.	**CO1**	**6**
B)	What is the nuclear chain reaction? Explain the importance of moderator and control rods in a nuclear reactor with respect to chain reaction	**CO1**	**6**
C)	What are the fossil fuels used for generation of conventional power? Explain in detail Steam power plant.	**CO1**	**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	How the wind mills are classified? Sketch the diagram of HAWT, and explain the function of its main components.	**CO2**	**6**
B)	What is Bio-mass? Write construction and working of bio-gas plant, with a neat diagram. Also write down the advantages of it.	**CO2**	**6**
C)	Define solar energy. What is flat plate collector? Describe its components with suitable sketch.	**CO2**	**6**
Q. 3	**Solve Any Two of the following.**		**12**
A)	What do you mean by energy conservation? Explain the measures to be taken to reduce the energy conservation in domestic activities. List any four measures**.**	**CO2**	**6**
B)	What do you understand by maximum energy efficiency in context with energy conservation principle? Discuss with a suitable example.	**CO1**	**6**
			6
Q.4	**Solve Any Two of the following.**		**12**
A)	Define Air Pollution. Write down the different classification of air pollution sources.	**CO3**	**6**
B)	Explain briefly effect of air pollution on human being and vegetation.	**CO1**	**6**
C)	What is radioactive pollution? What are its effects? How we can control Radioactive Pollution?	**CO3**	**6**
Q. 5	**Solve the following.**		**12**
A)	What are the main causes of water pollution? How can water pollution be controlled?		

B)	Explain the following terms: a. Thermal pollution b. Acid rain	**CO3**	**6**
C)	What are the various methods of safe disposal of solid wastes?	**CO3**	**6**

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